

# Factually: A Live, Wearable Fact-Checker

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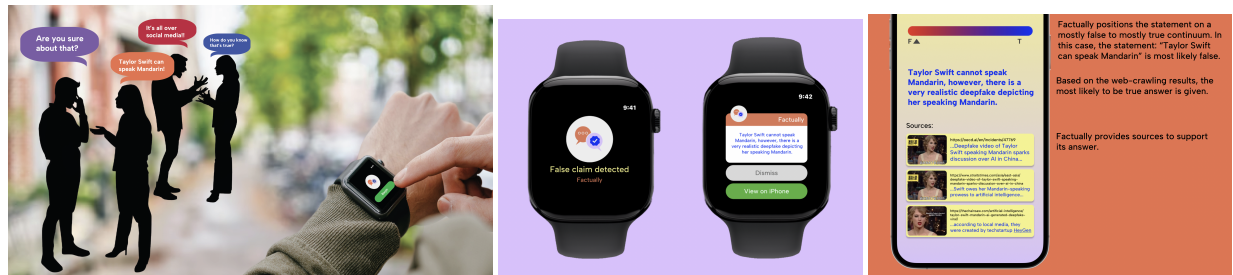


Figure 1: (a) A person using Factually in an everyday setting (b) Factually integrated with a smartwatch, (c) A companion mobile app.

## Abstract

We present *Factually*, a novel concept of a live, wearable, fact-checking system that extends users' ability to discern truth in real time. Integrated into devices like smartwatches or rings, Factually uses tactile vibrations to discreetly provide instant feedback on potential falsehoods. Unlike previous systems such as Wearable Reasoner or Facterverse, which focus on verbal feedback or broad applications, Factually emphasizes individual-level usability with non-intrusive, practical feedback. Through three illustrative scenarios, we demonstrate Factually's potential: extending perceptual abilities to detect misinformation in real time, enabling socially integrated and inconspicuous interactions, and fostering mindfulness by encouraging users to critically assess information. By addressing challenges of convenience, social dynamics, and misinformation, Factually showcases a step forward in empowering individuals to navigate an increasingly complex information landscape with enhanced critical thinking and truth-centered engagement.

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## CCS Concepts

• Human-centered computing → Haptic devices; Mobile devices.

## Keywords

Fact-checker, wearable devices, assistive augmentation

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## 1 Extended Abstract

Misinformation has become widespread in the digital era today, with bite-sized falsehoods often spreading rapidly through social media and everyday conversations. Identifying and addressing falsehoods mid-conversation is often inconvenient, socially awkward, and easily forgotten. To bridge this gap, we present *Factually*, a wearable live fact-checking system designed to augment individuals' ability to discern truth seamlessly within daily interactions.

In the realm of assistive augmentation [2], wearable devices have emerged as transformative tools, enhancing human cognition and redefining approaches to information processing. Factually advances this vision by operating as a discreet real-time assistant

integrated into smartwatches or rings. Unlike solutions like Wearable Reasoner, which use verbal feedback [1], Factually prioritizes usability with tactile feedback via vibrations, which are discreet. Similarly, while systems such as Factiveverse focus on live fact checking at scale [3], Factually emphasizes individual-level feedback to support instant, context-sensitive corrections.

To implement a proof-of-concept, we combined existing fact-checking mechanisms using large language models (LLMs) and web resources. While our prototype relies on current tools, the underlying fact-checking mechanism can be replaced with more advanced models. This work, however, focuses on wearability and assistive augmentation, showcasing Factually's seamless integration into users' lives and its cognitive enhancement potential.

Our video illustrates Factually's features through three key assistive augmentation [2] capabilities:

**Amplification of Cognitive Abilities:** Factually amplifies the user's ability to discern truth by providing real-time feedback on potentially false statements. For example, in one scenario, a child misidentifies a grandmother's medication as for high blood pressure, but Factually vibrates and alerts the user to its correct use, preventing misuse.

**Seamless Social Integration:** Factually's tactile feedback design leverages familiar, socially acceptable behaviors, such as glancing at a smartwatch or adjusting a ring, to enable inconspicuous use. In another scenario, users debunk a deepfake video of a celebrity speaking Mandarin without disrupting the flow of a casual lunch conversation. This illustrates Factually's ability to integrate into natural social interactions, fostering repeatable and practical usability in real-world settings.

**Encouraging Mindfulness and Truth-Centered Identities:** By subtly correcting user misconceptions in real time, Factually encourages critical thinking, mindfulness in speech, and opportunities for introspection. For instance, when a parent confidently but incorrectly asserts that dinosaurs lived with cavemen, Factually vibrates and provides the correct information. This moment not only guides the user toward accuracy but also encourages reflection on the source of their belief and a reassessment of how they share information. By supporting such self-correcting behaviors, Factually helps cultivate an identity rooted in truth, promoting mindful engagement with incoming information and fostering a deeper sense of responsibility in communication.

By empowering users with socially integrated, real-time truth discernment, Factually extends the boundaries of cognitive capabilities, while addressing key challenges in combating misinformation in everyday conversations. Through the video, we highlight Factually's potential to empower users to critically engage with information in real time, creating a foundation for future work in augmenting human cognition and resilience against misinformation.

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